

A Conditional Higher-Smoothness Upgrade for the $c_0(\Gamma)$ Embedding

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Status

This packet records a conditional reduction for a question in the 2024 paper of Hajek, Johannis, and Schlumprecht on point character and embeddings into $c_0(\Gamma)$ [1]. The source paper proves that, for the L_p and Banach-lattice classes in its Corollary 28, density below ω_ω gives a bi-Lipschitz embedding

$$\Phi : X \longrightarrow c_0(\Gamma)$$

whose scalar coordinates are C^1 smooth. Immediately after that corollary, the authors write that they do not know whether, for $p \geq 2$, the component functions can have higher smoothness.

The present packet does not claim an unconditional full answer. It isolates the exact smooth-approximation input that would upgrade the authors' C^1 conclusion to C^m , and verifies that this input is supplied for WCG spaces with a C^m smooth norm by Fry and Keener [2]. Thus the remaining point for human review is the formal differentiability-order upgrade of the smooth-coordinate c_0 embedding theorem used in the source paper.

Source Question

The source paper's relevant passage is on page 11.

the following corollary (note that any Lipschitz function from a subset of a metric space can be extended to the whole space with the same Lipschitz constant):

Corollary 28. *Let X be a subspace of $L_p(\mu)$ for some measure μ and $1 < p < \infty$, resp. of some super-reflexive Banach lattice with a (long) unconditional basis or a weak unit, with $\text{dens } X < \omega_\omega$. Then there is a bi-Lipschitz embedding $\Phi : X \rightarrow c_0(\Gamma)$ such that the component functions $f_\gamma \circ \Phi$ are C^1 -smooth, where f_γ are the canonical coordinate functionals on $c_0(\Gamma)$.*

We do not know whether in the corollary above in the case $p \geq 2$ the component functions can be of higher smoothness. Invoking [\[HJ\]](#) Theorems 7.79 and 7.86] we obtain a corollary on approximation of Lipschitz mappings:

Corollary 29. *Let X be a subspace of $L_p(\mu)$ for some measure μ and $1 < p < \infty$, resp. of some super-reflexive Banach lattice with a (long) unconditional basis or*

In text form, the question is:

We do not know whether in the corollary above in the case $p \geq 2$ the component functions can be of higher smoothness.

Conditional Dependency

The dependency below is the precise point where this packet remains conditional.

Lemma 1 (Formal C^m smooth-coordinate upgrade). *Let $m \in \mathbb{N} \cup \{\infty\}$. Suppose that X has the following Lipschitz C^m scalar approximation property: there is a constant A such that for every real-valued L -Lipschitz function f on X and every $\varepsilon > 0$ there exists a C^m smooth function $g : X \rightarrow \mathbb{R}$ with*

$$|f - g| < \varepsilon \quad \text{on } X, \quad \text{Lip}(g) \leq AL.$$

If X admits a bi-Lipschitz embedding into some $c_0(\Gamma)$, then X admits a bi-Lipschitz embedding into some $c_0(\Gamma')$ whose coordinate functions are C^m smooth.

This is not proved here from first principles. It is the formal C^m version of the C^1 smooth-coordinate theorem invoked by Hajek–Johanis–Schlumprecht through [1, Theorem 7.63] and Johanis’ theorem in their proof of Corollary 28. The construction should be order-free: the usual proof smooths scalar Lipschitz cut-off functions used in the c_0 embedding; if the approximants are C^m rather than merely C^1 , the estimates and the c_0 -valued support/threshold bookkeeping are unchanged. This last sentence is the human-check item.

Idea of the Proof

The source paper already supplies the metric part: for the stated L_p and Banach-lattice classes with dens $X < \omega_\omega$, X embeds bi-Lipschitzly into $c_0(\Gamma)$. Smoothness of the coordinates is then a separate scalar-approximation issue. Fry and Keener prove that on a WCG Banach space admitting a C^m smooth norm, Lipschitz real functions can be uniformly approximated by Lipschitz C^m smooth functions with controlled Lipschitz constant. Hence, once Lemma 1 is accepted, every such X inherits a $c_0(\Gamma')$ bi-Lipschitz embedding with C^m coordinate functions.

For subspaces of $L_p(\mu)$, $1 < p < \infty$, the structural hypotheses are available: these spaces are reflexive, hence WCG, and they inherit the usual C^m smooth norm in every order allowed by the scalar function $t \mapsto |t|^p$.

Conditional Theorem

Theorem 1. *Assume Lemma 1. Let $m \in \mathbb{N} \cup \{\infty\}$ and let X be a WCG Banach space which admits a C^m smooth norm. If X admits a bi-Lipschitz embedding into $c_0(\Gamma)$, then X admits a bi-Lipschitz embedding into some $c_0(\Gamma')$ whose coordinate functions are C^m smooth.*

Proof. By Fry–Keener [2], a WCG Banach space with a C^m smooth norm has the Lipschitz C^m scalar approximation property appearing in Lemma 1: Lipschitz real-valued functions can be uniformly approximated by C^m smooth Lipschitz functions with a Lipschitz constant controlled by a constant depending only on the space and the approximation scheme. Applying Lemma 1 to the given bi-Lipschitz $c_0(\Gamma)$ embedding yields the desired C^m smooth-coordinate embedding into another c_0 space. $\square$

Corollary 1 (Conditional L_p subcase). *Assume Lemma 1. Let X be a subspace of $L_p(\mu)$ with $1 < p < \infty$ and $\text{dens } X < \omega_\omega$. Let $m \in \mathbb{N}$. If $m < p$, then X admits a bi-Lipschitz embedding into some $c_0(\Gamma')$ whose component functions are C^m smooth. If p is an even integer, the same conclusion holds for every finite m .*

Proof. The source paper proves in Corollary 26 that such an X admits a bi-Lipschitz embedding into $c_0(\Gamma)$ when $\text{dens } X < \omega_\omega$. Since $1 < p < \infty$, the ambient $L_p(\mu)$ is reflexive, and therefore every closed subspace X is reflexive and WCG. For $m < p$ the usual L_p norm is C^m smooth away from zero in the standard Frechet sense; when p is an even integer the power function is polynomial and the usual norm gives all finite orders away from zero. Restricting the norm to X gives a C^m smooth norm on X . Theorem 1 completes the argument. $\square$

Verification Notes

The non-conditional parts of the argument are standard and local:

- $L_p(\mu)$ is reflexive for $1 < p < \infty$, so its closed subspaces are WCG.
- Smoothness of the L_p norm gives the asserted C^m orders, with the usual caveat at the origin handled by the definition of a smooth norm on $X \setminus \{0\}$.
- Fry–Keener’s arXiv abstract states the needed controlled Lipschitz C^p approximation theorem for WCG spaces admitting a C^p smooth norm.
- The source paper supplies the underlying $c_0(\Gamma)$ bi-Lipschitz embedding for subspaces of $L_p(\mu)$ of density below ω_ω .

The only nontrivial review item is Lemma 1. A direct check should revisit the proof of the C^1 coordinate-smoothing theorem cited in Corollary 28 and verify that replacing C^1 scalar approximants by C^m approximants preserves the support/threshold construction ensuring that the map is c_0 -valued and bi-Lipschitz.

Limitations

This packet does not solve the full higher-smoothness question. It is conditional on Lemma 1; it does not cover arbitrary super-reflexive spaces, arbitrary WCG finite-cotype spaces, or all orders of smoothness for non-even p . It also does not address the two broader problems at the end of the source paper: the general super-reflexive/WCG embedding problem and the question whether a bi-Lipschitz embedding of B_X into Y forces one of X into Y .

Novelty and Literature Check

The run indexes were searched for arXiv:2401.00831, point-character keywords, smooth-coordinate keywords, and higher-smoothness phrases. No existing packet or attempt was found. Web searches for exact phrases from the source question and for the source title plus “higher smoothness” found the source paper itself and no later explicit answer. Fry–Keener [2] is used as a supporting smooth-approximation theorem rather than as an explicit answer to the source question.

Human Review Recommendation

Review Lemma 1 first. If the C^1 smooth-coordinate embedding theorem cited by the source paper is indeed formal in the smoothness class once controlled Lipschitz scalar approximation is available, then Corollary 1 should be promotable from conditional to a partial answer to the higher-smoothness question.

References

- [1] P. Hajek, M. Johanis, and T. Schlumprecht, *Remarks on the point character of Banach spaces and non-linear embeddings into $c_0(\Gamma)$* , arXiv:2401.00831, 2024.
- [2] R. Fry and L. Keener, *Approximation of Lipschitz functions by Lipschitz, C^p smooth functions on weakly compactly generated Banach spaces*, arXiv:0907.0241, 2009.